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To the Editors, Annales de la Fondation Louis de Broglie | inst.louisdebroglie@free.fr
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Dear Editors,

I submit for your consideration the article *Magnetism in RealQM: Real Charge Currents, Diamagnetism, and the Geometry of Spin*. It is a companion to my earlier submission on RealQM — quantum mechanics formulated as N real charge densities on non-overlapping domains in ordinary three-dimensional space, in the de Broglie–Schrödinger tradition — and it applies that real-space ontology to the one arena where the reality of the wave function is most sharply at stake: magnetism and spin.

The article makes a graded, and deliberately honest, case. *Orbital magnetism* is shown to be real continuum mechanics: a magnetic moment is a real circulating current $\mathbf{j} = \rho \mathbf{v}$ carrying real angular momentum, and the complex wave function $\psi = \sqrt{\rho} e^{iS}$ merely bundles the two real fields $(\rho, \mathbf{v})$ that continuum mechanics has always carried without any recourse to complex numbers; the sole irreducibly quantum input is the *integer* winding number, fixed by single-valuedness exactly as Onsager and Feynman fixed the quantisation of superfluid vortices. *Diamagnetism* then furnishes a quantitative test: the Langevin susceptibility $\chi \propto -\langle r^2 \rangle$ is read directly from the RealQM density, and a radial self-consistent field reproduces the measured molar susceptibility of helium to $0.99\times$, with neon and argon correct in order and trend and the shortfall diagnosed.

Spin is treated as the single residue it is, and I have tried to be scrupulous about what is and is not established. The article argues that spin need not be posited as a primitive in RealQM: its exclusion role is already geometry (non-overlapping domains), its mechanical angular momentum is already the real circulation of charge (so the Einstein–de Haas effect is native rather than an obstacle), and the quantisation of its magnitude is the same topological closure that quantises the orbital moment — applied to a spinorial orientation, whose 4π (SU(2)) periodicity yields $g = 2$. By Lévy-Leblond’s theorem this factor is non-relativistic geometry, not a relativistic import, and it is delivered for one electron by a $\boldsymbol{\sigma} \cdot \mathbf{p}$ structure with $g = 2$ emerging rather than inserted. I state equally plainly what remains open: that the half-integer closure is exhibited but not yet forced by the domain geometry, and that closed-shell pairing is *not* supplied by the spin-blind non-overlap geometry and requires an added exchange term — a genuine limitation, reported as such.

I submit to the *Annales* because this is foundational work in the de Broglie–Schrödinger spirit: it asks how much of magnetism follows from taking the wave function to be a real charge density in three-dimensional space, gives a quantitative and testable answer, and marks its own boundary honestly rather than overclaiming. The numerical results are reproducible; the underlying framework runs in a browser and is documented in a public Gallery.

Thank you for your consideration.

Sincerely,
Claes Johnson